- *Programmatically*—If neither of the preceding is an option, you can add seed nodes using an implementation that you write yourself.

Because this is an introductory chapter about clustering, we only cover the definition of seed nodes through configuration. In this setting, you specify a list of seed nodes with a host and a port for each of them. The first one in the list plays a special role in forming the cluster; other seeds can join the cluster only if it is `Up`. This restriction was introduced to prevent individual clusters from forming while the seed nodes are starting up.

The seed nodes can all boot independently as long as the first seed node is started at a given time. Other nodes that are not seed nodes can join the cluster through one of the seed nodes after the first seed node is started and `Up`. Once the first seed node has formed the cluster and set all other seed nodes to `Up`, it can safely leave the cluster and pass its leadership to the next seed node in the list: that is, the one with the lowest IP address. Figure 8.4 shows the nodes in the cluster for the word-counting application.